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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: B.E / B.Tech
Course Code: MLA02
Course Name: FUNDAMENTALS OF MACHINE LEARNING
Duration: 1:30 Hr
Date : 2-11-2026

Max Marks: 50 Marks

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at $x = 6$. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at $x = 6$.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing (α = 1). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(6) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(4) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL4)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(3) (CO2) (BL5)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(4) (CO2) (BL4)

(3) (CO2) (BL4)

AdaBoost starts with five training samples of equal weight. The first weak learner misclassifies samples 2 and 5. Compute the weighted error, the learner weight α, the un-normalised and normalised weights after the update, and analyse how boosting changes the focus of the next learner.

(4) (CO2) (BL4)

Task 11. Two classes share the covariance matrix $\Sigma = [[2, 0.5],[0.5, 1]]$ and have means $\mu_1 = (1, 2)^T$ and $\mu_2 = (3, 1)^T$. (a) Derive the discriminant direction $w = \Sigma^{-1}(\mu_2 - \mu_1)$ and the decision threshold for equal priors. (b) Classify $x = (2.5, 1)$. (c) Evaluate whether the decision changes if the priors are $P(\omega_1) = 0.8$ and $P(\omega_2) = 0.2$.

(4) (CO2) (BL4)

Task 12. A bank evaluates three classifiers on 1,000 test loans (100 actual defaulters). Default is the positive class.

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

A missed defaulter (FN) costs ₹50,000 and a wrongly rejected good applicant (FP) costs ₹5,000.

(a) Compute accuracy, precision, recall and F1-score for each model. [3]

(b) Compute the total expected cost of each model. [2]

(c) Critically evaluate which model should be deployed and explain why accuracy and F1 lead to a different choice than cost. [2]

(d) Propose two modelling or threshold-tuning steps to improve the chosen model and one validation safeguard against over-fitting. [1]

(8) (CO2) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech	**Course Code**	**MLA02**
Course Name	**Fundamentals of Machine learning.**	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	**50**

Register No.: 192461010

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	5
Task 4	4	4
Task 5	3	3
Task 6	3	3
Task 7	4	3
Task 8	4	4
Task 9	3	3
Task 10	4	4
Task 11	4	4
Task 12	8	8
TOTAL	**50**	48

Total in words:

linear regression using OLS.

Given:

$x = 1, 2, 3, 4, 5$

$y = 2.1, 3.9, 6.2, 7.8, 10.1$

a) fit the regression equation.

$y = \omega_0 + \omega_1 x$

$\bar{x} = 3, \quad \bar{y} = 6.02$

$S_{xx} = 10, \quad S_{xy} = 19.9$

$\omega_1 = \frac{19.9}{10} = 1.99$

$\omega_0 = 6.02 - (1.99)(3) = 0.05$

Regression equation:

$y = 0.05 + 1.99x$

b) predict at $x = 6$

$y = 0.05 + 1.99(6)$

$y = 11.99$

c) Error calculations.

$RSS = \Sigma (y - \hat{y})^2 = 0.107$

$MSE = \frac{0.107}{5} = 0.0214$

$R^2 = 1 - \frac{0.107}{39.708}$

$R^2 = 0.9973.$

d) Reliability :- The model has a very high R^2 (99.73%) indicating a good fit. However, $x = 6$, is outside the training range, so extrapolation may be less reliable.

2. Ridge Regression.

Given:

$S_{xx} = 10$, $S_{xy} = 19.9$, $\lambda = 5$

Ridge slope:

$$w = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{19.9}{10+5} = \frac{19.9}{15}$$

$$w = 1.3267$$

Least-squares slope:

$$w_{LS} = \frac{19.9}{10} = 1.99$$

percentage shrinkage:

$$= \frac{1.99 - 1.3267}{1.99} \times 100$$

$$= 33.33\%$$

conclusion :- Increasing λ increases bias & decreases variance, reducing overfitting.

3. Naive Bayes classification.

Given:

10 emails:

spam = 5

Ham = 5

a) priors & conditional probabilities

$$P(\text{spam}) = P(\text{Ham}) = \frac{5}{10} = 0.5$$

atores

	spam	Ham.
offer = Y	4/5	1/5
offer = N	1/5	4/5
cink = y	4/5	1/5
Cink = N	1/5	4/5

b) offer = y , Link = y

$P(\text{spam} \mid y, y) \propto 0.5(0.8)(0.8) = 0.32$

$P(\text{Ham} \mid y, y) \propto 0.5(0.2)(0.2) = 0.02$

$P(\text{spam} \mid y, y) = \frac{0.32}{0.34} = 94.12\%$

class = spam.

c) Laplace smoothing ($\alpha = 1$)

$P(\text{offer} = y \mid \text{spam}) = \frac{4+1}{5+2} = \frac{5}{7}$

$P(\text{Link} = y \mid \text{spam}) = 5/7$

for Ham, both probabilities are 2/7

$P(\text{spam} \mid y, y) = \frac{25}{25+4} = \frac{25}{29} = 86.21\%$

d) offer = N , Link = N (without smoothing)

$P(\text{spam} \mid N, N) = \frac{0.5(0.2)(0.2)}{0.5(0.2)(0.2) + 0.5(0.8)(0.8)}$

$= \frac{0.02}{0.02 + 0.32} = \frac{0.02}{0.34}$

$\therefore 5.88\%$

class = Ham.

with Laplace smoothing :

$$P(\text{spam} \mid N,N) = \frac{4}{4+25} = \frac{4}{29} = 13.79\%.$$

Conclusion :- Naive Bayes assumes conditional independance, Laplace smoothing prevents zero probabilities & improves estimation.

4. Logistic Regression - Gradient Descent

Given :

$$w = (0,0), \quad b = 0, \quad \eta = 0.5$$

Initial predictions :

$$P_1 = P_2 = \sigma(0) = 0.5$$

Gradient of weights :

$$\nabla w = \frac{1}{2}[(-0.5)(1,2) + (0.5)(2,1)]$$

$$\nabla w = (0.25, -0.25)$$

Bias gradient :

$$\nabla b = \frac{1}{2}(-0.5 + 0.5) = 0$$

Updated parameters :

$$w_1 = 0 - 0.5\,(0.25) = -0.125$$

$$w_2 = 0 - 0.5\,(-0.25) = 0.125$$

$$b = 0.$$

Updated predictions :

$$P_1 = \sigma(0.125) = 0.5312$$

$$P_2 = \sigma(-0.125) = 0.4688$$

Updated loss :

$$L = 0.6931$$

$$L = -\ln(0.5312)$$

$$\therefore L \approx 0.63$$

tial loss :

$L = 0.6931$

updated loss :

$l = -\ln(0.5312)$

$\therefore l \approx 0.631$

Decision tree - information gain .

Given ;

- Total samples = 10
- yes = 6, NO = 4
- Attribute A_1 = 4 yes, 4 NO
- Attribute A_2 = 6 yes, 0 NO.

i Entropy of parent

$H(S) = -0.6\log_2(0.6) - 0.4\log_2(0.4)$

$H(S) = 0.971$

ii information gain

Both child nodes are pure.

$H(S|A) = 0$

$IG(A) = 0.971 - 0$

$IG = 0.971$

iii Gini impurity

$Gini(S) = 1 - (0.6)^2 - (0.4)^2$

$= 0.48$

children Gini = 0

$Gini\ \Psi = 0.48 - 0$

$= 0.48$

6. cost - complexity pruning.

Given:

- subtree training errors = 3
- Leaf errors = 8
- subtree leaves = 4
- single leaf = 1

$$R_\alpha(T) = 3 + 4\alpha$$

$$R_\alpha(\text{leaf}) = 8 + \alpha$$

At equality:

$$3 + 4\alpha = 8 + \alpha$$

$$3\alpha = 5$$

$$\alpha = 1.667$$

for $\alpha = 0.01$ & $\alpha = 0.02$.

Both are less than 1.667.

7. Neural Network - Back propagation.

Given;

$x = (1, 2)$, $w_1 = 0.4$, $w_2 = 0.3$

$v = 0.5$, $t = 1$, $\eta = 0.5$

i) forward propagation.

Hidden output:

$$h = \sigma(0.4 + 0.3(2))$$

$$h = \sigma(1) = 0.7311$$

output:

$$\hat{y} = \sigma(0.5(0.7311))$$

$$\hat{y} = 0.5904$$

$$\epsilon = 1/2\,(1-0.5904)^2$$

$$\epsilon = 0.0839$$

(iii) Back propagation.

Output error :

$$\delta_0 = (0.5904 - 1)(0.5904)(1-0.5904)$$

$$\delta = -0.0991$$

Hidden error :

$$\delta_h = \delta_0 (0.5)(0.7311)(1-0.7311)$$

$$\delta_h = -0.00974$$

Gradients :

$$\frac{\partial \epsilon}{\partial u} = -0.07242$$

$$\frac{\partial \epsilon}{\partial w_1} = -0.00974$$

$$\frac{\partial \epsilon}{\partial w_2} = -0.01948$$

(iv) Updated weights.

$$u = 0.5 - 0.5(-0.07242) = 0.5362$$

$$w_1 = 0.4 - 0.5(-0.00974) = 0.4049$$

$$w_2 = 0.3 - 0.5(-0.01948) = 0.3097.$$

New loss :

$$\epsilon \approx 0.0811$$

Conclusion : Loss ↓ after back propagation.

8. Support vector machine (SVM)

Given:

$w = (0.5, 0.5)$, $b = -2$

Decision function:

$$f(x) = 0.5x_1 + 0.5x_2 - 2$$

a) verify support vectors.

for (3,3):

$$f(3,3) = 1$$

for (1,1):

$$f(1,1) = -1$$

Margin width:

$$\frac{2}{\|w\|} = \frac{2}{\sqrt{0.5}} = 2.828$$

b) classify points.

for (4,2):

$$f = 1 \Rightarrow +1$$

for (1,2):

$$f = (-0.5) = -1$$

Distance from boundary:

$$d(4,2) = \frac{1}{\sqrt{0.5}} = 1.414$$

$$d(1,2) = \frac{0.5}{\sqrt{0.5}} = 0.707$$

c) Hinge loss

$$L = \max(0, 1 - y f(x))$$

$L = \max(0, 1-0.5) = 0.5$

for (2,2), $y = +1$:

$f = 0.$

$L = \max(0,1) = 1$

Conclusion: Both points violate the margin.

9. Ensemble learning.

Given:

$P = 0.70, q = 0.30$

i Three classifiers.

$P = \binom{3}{2}(0.7)^2(0.3) + (0.7)^3$

$= 0.441 + 0.343$

$= 78.4\%.$

ii, five classifiers.

$P = \sum_{k=3}^{5} \binom{5}{k}(0.7)^k (0.3)^{5-k}$

$= 0.343 + 0.36015 + 0.16807$

$= 87.12\%.$

10. AdaBoost

Given:

- Total samples = 5
- Initial weight = 0.2 each
- Misclassified samples = 2 & 5

i. weighted error.

$$\epsilon = 0.2 + 0.2 = 0.4$$

ii. learner weight

$$\alpha = \frac{1}{2} \ln \frac{1-\epsilon}{\epsilon}$$

$$= \frac{1}{2} \ln \frac{0.6}{0.4}$$

$$\boxed{\alpha = 0.2027}$$

iii. weight update.

Misclassified samples:

$$0.2e^{0.2027} = 0.24495$$

correctly classified samples:

$$0.2e^{-0.2027} = 0.16330$$

Normalization factor:

$$z = 0.9788$$

Normalized weights:

samples	New weight.
1	0.1667
2	0.2500
3	0.1667
4	0.1667
5	0.2500

11. Linear Discriminant Analysis (LDA)

Given:

$$\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$\mu_1 = (1,2)$, $\mu_2 = (3,1)$

a) Discriminant direction,

$$w = \Sigma^{-1}(\mu_1 - \mu_2)$$

$$\Sigma^{-1} = \frac{1}{1.75}\begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}$$

$$w = (-1.4286, 1.7143)$$

b) classify $x = (2.5, 1)$

equal-prior decision boundary:

$$w^T x = \frac{w^T(\mu_1 + \mu_2)}{2}$$

threshold:

$$= -0.2857$$

for x:

$$w^T x = -1.8571$$

since: $-1.8571 < -0.2857$

class = ω_2

c) changed priors.

$$P(\omega_1) = 0.8, \quad P(\omega_2) = 0.2$$

New decision score:

$$-1.8571 - (-0.2857) + \ln(4)$$

$$= -0.1857$$

since it is -ve:

class = ω_2

12. classified evaluation.

Given total = 1000 loans.

a) Accuracy, precision, recall & f1-score.

Formulas:

$$\text{Accuracy} = \frac{TP + TN}{1000}$$

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

$$f_1 = \frac{2TP}{2TP + FP + FN}$$

Model A.

$$\text{Accuracy} = \frac{70 + 810}{1000} = 88\%$$

$$\text{precision} = \frac{70}{160} = 43.75\%$$

$$\text{recall} = \frac{70}{100} = 70\%$$

$$f_1 = \frac{140}{260} = 53.85\%$$

Model B:

$$\text{Accuracy} = \frac{85 + 720}{1000} = 80.5\%$$

$$\text{precision} = \frac{85}{265} = 32.08\%$$

$$\text{recall} = \frac{85}{100} = 85\%$$

$$f_1 = \frac{170}{350} = 48.57\%$$

del C :

$$\text{Accuracy} = \frac{60+970}{1000} = 93\%$$

$$\text{precision} = \frac{60}{90} = 66.67\%$$

$$\text{Recall} = \frac{60}{100} = 60\%$$

$$f_1 = \frac{120}{190} = 63.16\%$$

Comparison table

Model	Accuracy	precision	Recall	f_1
A	88%	43.75%	70%	53.85%
B	90.5%	32.08%	85%	48.87%
C	93%	66.67%	60%	63.16%

b) Total expected cost

Given :

FN cost = ₹ 10,000

FP cost = ₹ 1000

cost = FN (10000) + FP (1000)

Model A :

30 (10000) + 90 (1000)

= ₹ 19,10,000

Model B :

15 (10000) + 190 (1000)

= ₹ 16,50,000

Model C :

40(10000) + 30(8000)

= ₹ 21,50,000

c) Best model selection.

Model B (lowest cost)

Model C has the highest accuracy & f1-score, but Model B has the lowest total financial cost because it misses fewer defaulters.

d) Improvement techniques.

1. Apply class weighting or oversampling to improve minority-class detection.
2. tune the classification threshold to reduce false negative.

validation safeguard : Use stratified cross-validation to prevent overfitting.